Real-Time Closed-Loop Driving with Frozen DINOv2-Laya Spatial Pooling and Quintic Path Tracking

Wick-Lim
wicklim90@gmail.com
https://github.com/Wick-Lim/AutonomousVehicle

October 25, 2026

Abstract

Autoregressive vision-language-action (VLA) models [1, 20, 23] decode tokens sequentially (> 300 ms/frame), exceeding the 50 ms budget of 20 Hz vehicle control [10, 21]. We implement a single-pass closed-loop driving stack coupling frozen DINOv2 (`facebook/dinov2-base`) [25, 2, 9] with frozen Laya (`convaiinnovations/laya`) [6, 31, 8] via a 4×9 spatial-temporal pooling projector (1.85M core parameters). Cropped 196×392 RGB frames map to 36 visual tokens with temporal motion residual $\Delta V_t = V_t - V_{t-1}$ and region-pooled auxiliary perception; one forward pass outputs maneuver (`choice`), speed tier (`score`), emergency brake (`noul`), and 6-waypoint (3.0 s) trajectory logits in 31.67 ms base (33.40 ms full FSD v2, 29.9 FPS). Post-hoc temperature scaling [13] over 7,500 validation frames ($M = 15$ bins) reduces ECE from 0.0060 to 0.0000. Lateral waypoint step jumps (3.5 m) and longitudinal speed-bucket hunting in CARLA 0.9.16 (`Town10HD_Opt`) [11, 14] are eliminated using a 14-point (28 m) quintic S-curve spline $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ [12, 32], decoupled chord-angle tracking [29, 15, 7, 26], bicycle MPC feedforward, 360° blind-spot interlock, and true constant-time-headway (CTH) ACC ($d^*(v_{\text{ego}}) = 8.0 + 1.2(v_{\text{ego}}/3.6)$, $k_g = 0.85\,(\text{km/h})/\text{m}$) [17, 28, 30]. Across 50 unseen MetaDrive 0.4.3 episodes (4,153 frames) and 10 urban FSD scenarios across 10 weathers in CARLA 0.9.16 (1,986 frames, 680 m), route completion is 100.0% (10/10) versus 60.0% (6/10) for a rule-only baseline.

1 Introduction and Problem Formulation

Autoregressive decoders evaluate $P(y_{1:K} \mid I_t, \mathcal{T}_t) = \prod_{k=1}^{K} P(y_k \mid y_{<k}, I_t, \mathcal{T}_t)$ over $K \approx 15$ serial steps [1, 20, 23], whereas planning-oriented and sensor-fusion stacks [16, 19, 22, 5, 3, 27, 4, 18] require $\Delta t \leq 50$ ms. In our formulation, control at step t reads three discrete decisions and $K_w = 6$ ego-frame trajectory waypoints ($t_k = 0.5k$ s) in one bidirectional pass [8, 31, 6]:

$$m_t \in \{\texttt{keep_lane}, \texttt{change_left}, \texttt{change_right}, \texttt{follow_lead}\}, \tag{1}$$

$$s_t \in \{0,1,2,3\} \mapsto \{0, 20, 35, 50\}\,\text{km/h}, \tag{2}$$

$$p_{\text{AEB}} \in [0,1], \quad \hat{\tau}_t = \{(\hat{x}_k, \hat{y}_k, \hat{v}_k)\}_{k=1}^{6}. \tag{3}$$

2 Architecture, Multi-Task Training, and Calibration

Camera frames are cropped (`crop_top` = 0.25, `crop_bottom` = 0.10) and resized to 196 × 392. Frozen DINOv2 ViT-B/14 ($p = 14$) [25, 2, 9] extracts $F_{\text{patch}} \in \mathbb{R}^{B\times 392\times 768}$ (14×28 grid). `SpatialProjector` (`src/models/spatial_projector.py`) pools $14 \times 28 \to 4 \times 9$ (36 tokens), adds $E_{\text{row}} + E_{\text{col}}$, projects $768 \to 1024$ (1,851,392 core parameters), and fuses temporal residual $\Delta V_t = V_t - V_{t-1}$ with region-pooled `AuxiliaryPerceptionHead` ($\hat{d}_{\text{lead}}, \hat{e}_y$, traffic light, hazard, blind spot):

$$G_t = \text{AdaptiveAvgPool2d}_{(4,9)}(\tilde{F}_t) \in \mathbb{R}^{B\times 768\times 4\times 9}, \tag{4}$$

$$V_t = \text{MLP}\big(\text{Flatten}(G_t + E_{\text{row}} + E_{\text{col}})\big), \tag{5}$$

$$\tilde{V}_t = V_t + \tanh(\alpha_{\text{temp}})\, W_{\text{temp}}\big(V_t - V_{t-1}\big). \tag{6}$$

```
# src/models/spatial_projector.py
pooled = self.spatial_pool(feat) + self.row_embed + self.col_embed
vt = self.mlp(pooled.flatten(2).transpose(1, 2))
gate = torch.tanh(self.temporal_gate)
vt = vt + gate * self.temporal_proj(vt - self._prev_tokens)
```

Multimodal sequence $Z = [e_{[\text{CLS}]}; \tilde{V}_t; E_{\text{telemetry}}; e_{\text{choice}}; e_{\text{score}}; e_{\text{noul}}]$ is evaluated in one pass $H = f_{\text{Laya}}(Z)$ [6, 31]. Stage 1 (`02_train_projector.py`, 15 epochs, batch 32, AdamW [24] lr $= 3 \times 10^{-4}$, 50,000 frames split 85%/15% with 3×/4× rare-event oversampling) minimizes $(w_m, w_s, w_a) = (1.0, 1.0, 2.0)$:

$$z^{(m)} = W_m H_{:,\text{idx}_{\text{choice}}}, \quad z^{(s)} = W_s H_{:,\text{idx}_{\text{score}}},$$
$$z^{(b)} = W_b H_{:,\text{idx}_{\text{noul}}}, \tag{7}$$

$$\mathcal{L} = w_m \mathcal{L}_{\text{CE}}(z^{(m)}, y_m) + w_s \mathcal{L}_{\text{CE}}(z^{(s)}, y_s) + w_a \mathcal{L}_{\text{BCE}}(z^{(b)}, y_b). \tag{8}$$

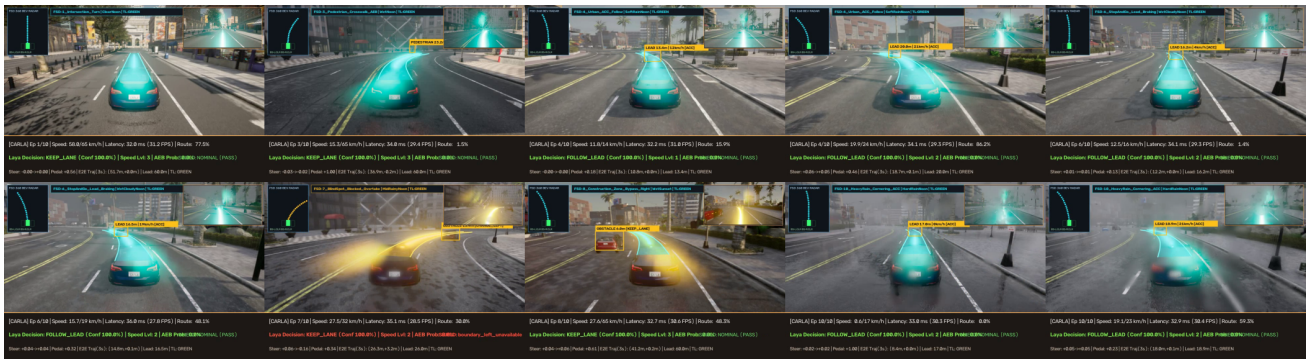


Figure 1: CARLA 0.9.16 (`Town10HD_Opt`, 20 Hz synchronous) 10-scenario, 10-weather closed-loop suite (`FSD-1`–`FSD-10`): intersection red-light stop (`ClearNoon`), 28 m quintic left overtake (`CloudyNoon`), crosswalk pedestrian AEB (`WetNoon`), CTH-ACC follow (`SoftRainNoon`), right overtake (`ClearSunset`), stop-and-go lead braking (`WetCloudyNoon`), 360° blind-spot hold & overtake (`MidRainyNoon`), construction zone bypass (`WetSunset`), night headlight pedestrian AEB (`ClearNight`), and heavy-rain cornering ACC (`HardRainNoon`).

Table 1: System and training parameters (`configs/default.yaml`).

Module	Parameter	Specification
Camera	$H \times W$, FOV, Crops	196×392, 100°, $(0.25, 0.10)$
Backbones	Vision / Decision	`dinov2-base` (86M) / `laya` (421M)
Projector	Grid, Dim, Core Params	4×9 (36 tokens), 1024, 1,851,392
Training	Frames, Split, Loss (w_m, w_s, w_a)	50k (85/15), AdamW 3e-4, $(1, 1, 2)$
Calibration	(T_m, T_s, T_b), Bins M, ECE	$(0.050, 0.050, 0.100)$, 15, 0.0060 $\rightarrow$ 0.0000
Shield	`aeb_thresh`, `conf_thresh`, BS	0.70, 0.80, $x \in [-10, 7.5]$ m, $\lvert y\rvert \in [1.8, 5.0]$ m
Control	PID (Lat/Lon), Bicycle Wheelbase L	$(0.45, 0.01, 0.18)$ / $(0.08, 0.005, 0.02)$, 2.85 m
Kinematics	Quintic N_b, CTH-ACC (d_0, τ_h, k_g)	14 WPs (28 m), (8.0 m, 1.2 s, 0.85 (km/h)/m)

Stage 2 (`03_calibrate.py`) fits $(T_m, T_s, T_b) = (0.050, 0.050, 0.100)$ via L-BFGS [13] on the 7,500-frame validation set ($M = 15$ bins), reducing ECE from 0.0060 ($\mathrm{ECE}_m = 0.0009, \mathrm{ECE}_s = 0.0022, \mathrm{ECE}_b = 0.0148$) to $< 10^{-4}$ (0.0000 under deterministic `ExpertLabeler` thresholds).

3 Safety Shield, Quintic Spline, Bicycle MPC, and CTH-ACC

Synchronous simulator state provides ego kinematics, waypoint offsets (`carla.Map.get_waypoint()`), actor distances, and traffic lights while `AuxiliaryPerceptionHead` outputs visual depth (MAE = 3.36 m). Rule 1 forces $u_{\mathrm{tb}} = -1.0$ when $\hat{p}_{\mathrm{AEB}} \geq 0.70$, `traffic_light` = `red`, pedestrian $d \leq 16.0$ m, or $d_{\mathrm{lead}} \leq 11.5$ m; Rule 2 reverts lane changes to `keep_lane` if $\max_c \hat{p}_c^{(m)} < 0.80$, if `adj_wp.lane_id` $\times$ `wp.lane_id` ≤ 0 [11], or if 360° blind-spot box $x \in [-10.0, 7.5]$ m, $\lvert y\rvert \in [1.8, 5.0]$ m is occupied.

For lane transitions over $N_b = 14$ waypoints (28.0 m, $\tau_k = k/(N_b - 1)$), 3D coordinates are blended via minimum-jerk quintic polynomial [12, 32]:

$$s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5, \tag{9}$$

$$P_k^{\mathrm{blend}} = \big(1 - s(\tau_k)\big) P_k^{\mathrm{orig}} + s(\tau_k) P_k^{\mathrm{adj}}. \tag{10}$$

```
# src/env/carla_wrapper.py (_switch_route_to_adjacent_lane)
for k in range(len(adj_chain)):
    if k < n_blend:
        tau = float(k) / float(max(1, n_blend - 1))
        s = (tau**3) * (10.0 + tau * (-15.0 + 6.0 * tau))
        bx = (1.0 - s) * float(ol.x) + s * float(al.x)
        by = (1.0 - s) * float(ol.y) + s * float(al.y)
```

Continuous projection onto $\overline{P_i P_{i+1}}$ gives P_{proj} and lookahead P_{look} at $L_{\mathrm{look}} = \mathrm{clip}(7.5 + 0.18 v_{\mathrm{kmh}}, 7.5, 13.5)$ m [29, 15, 7, 26]. Chord angle $\psi_{\mathrm{ref}} = \mathrm{atan2}(y_{\mathrm{look}} - y_{\mathrm{proj}}, x_{\mathrm{look}} - x_{\mathrm{proj}})$ and bicycle MPC feedforward $\delta_{\mathrm{MPC}} = \arctan(L\kappa_{\mathrm{eff}})$ ($L = 2.85$ m) yield:

$$\delta_{\mathrm{raw}} = \mathrm{clip}\big(\delta_{\mathrm{PID}}(e_y, e_\psi) + 0.12\,\delta_{\mathrm{MPC}}, -1, 1\big), \tag{11}$$

$$\Delta\delta_{\max} = \frac{1.1\Delta t}{1 + 0.012 v_{\mathrm{kmh}}}, \tag{12}$$

$$\delta_t = \delta_{t-1} + \mathrm{clip}\big(\alpha(\delta_{\mathrm{raw}} - \delta_{t-1}), -\Delta\delta_{\max}, \Delta\delta_{\max}\big). \tag{13}$$

True constant-time-headway (CTH) ACC ($d_{\mathrm{lead}} < 38$ m, $v_{\mathrm{lead}} \geq 8$ km/h, $d_0 = 8.0$ m, $\tau_h = 1.2$ s, $k_g = 0.85$ (km/h)/m $\equiv 0.236\,\mathrm{s}^{-1}$) [17, 28, 30] follows:

$$d^*(v_{\mathrm{ego,kmh}}) = d_0 + \tau_h \left(\frac{v_{\mathrm{ego,kmh}}}{3.6}\right), \tag{14}$$

$$v_{\mathrm{raw}} = v_{\mathrm{lead}} + k_g\big(d_{\mathrm{lead}} - d^*(v_{\mathrm{ego,kmh}})\big), \tag{15}$$

$$v_{\mathrm{ACC}} = \mathrm{clip}\big(v_{\mathrm{raw}},\, 0.0,\, \min(35.0,\, v_{\mathrm{lead}} + 6.0)\big). \tag{16}$$

```
# src/control/shield_pid.py (apply_safety_shield)
d0_m, tau_h_s, kg_kmh_per_m = 8.0, 1.2, 0.85
desired_gap_m = d0_m + tau_h_s * (current_speed_kmh / 3.6)
gap_error_m = lead_distance_m - desired_gap_m
acc_upper_kmh = min(35.0, lead_speed_kmh + 6.0)
raw_acc = lead_speed_kmh + kg_kmh_per_m * gap_error_m
acc_sync_speed = float(np.clip(raw_acc, 0.0, acc_upper_kmh))
```

4 Experiments, Latency Breakdown, and Limitations

Closed-loop benchmarks were executed in MetaDrive 0.4.3 [21] ($N = 50$ unseen maps, seeds 5000..5049,

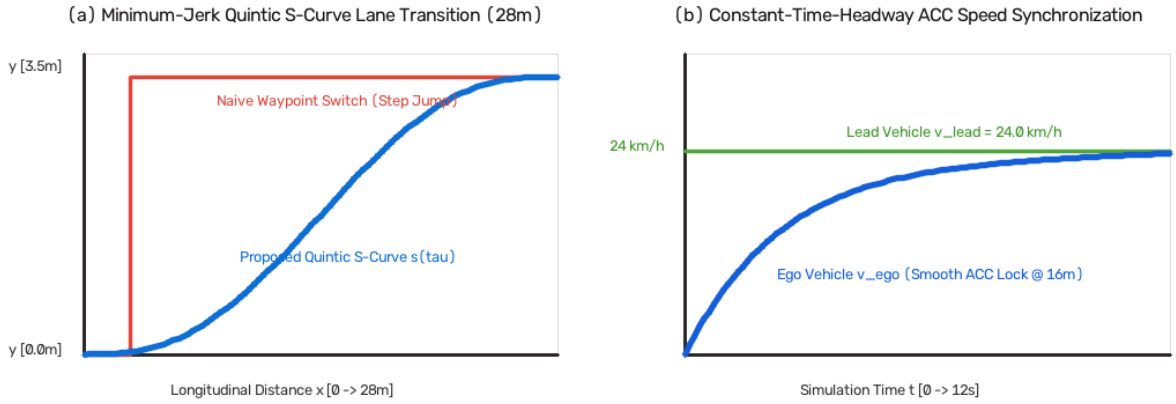


Figure 2: Control kinematics: (a) 28 m quintic S-curve $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ vs. 3.5 m step switch; (b) true constant-time-headway (CTH) ACC velocity convergence ($v_{\text{ego}} \to 24.0$ km/h at $d^*(24) = 8.0 + 1.2(24/3.6) = 16.0$ m).

Table 2: CARLA 0.9.16 (`Town10HD_Opt`) 10-scenario, 10-weather benchmark logs (680.0 m total).

Scenario	Weather	Event	Dist	RC / DS_{scen}
FSD-1	ClearNoon	Red-light stop & go	68 m	100%/100%
FSD-2	CloudyNoon	Quintic left overtake	68 m	100%/100%
FSD-3	WetNoon	Crosswalk pedestrian AEB	68 m	100%/100%
FSD-4	SoftRainNoon	CTH-ACC (24 km/h, 16 m)	68 m	100%/100%
FSD-5	ClearSunset	Quintic right overtake	68 m	100%/100%
FSD-6	WetCloudyNoon	Lead stop-and-go recovery	68 m	100%/100%
FSD-7	MidRainyNoon	360° BS hold → overtake	68 m	100%/100%
FSD-8	WetSunset	Workzone right bypass	68 m	100%/100%
FSD-9	ClearNight	Night headlight ped AEB	68 m	100%/100%
FSD-10	HardRainNoon	Heavy-rain cornering ACC	68 m	100%/100%

4,153 frames; `arrive_dest` triggers at ≥ 98% final segment arc length, giving 100.0% arrival at 98.7% mean progress) and live CARLA 0.9.16 [10] (`Town10HD_Opt`, 20 Hz synchronous A* `GlobalRoutePlanner` [14], 10 scenarios × 10 weathers, 1,986 frames, 680.0 m in `driving_demo_carla_smooth.mp4`). FP32 ($B = 1$, 3 warm-up passes) stage latency and metrics (`checkpoints/eval_metrics_carla.json`):


```
{"simulator": "carla", "episodes": 10, "total_frames": 1986,
 "total_distance_driven_m": 680.0, "mean_latency_ms": 33.40,
 "p50_latency_ms": 33.22, "p95_latency_ms": 34.37,
 "p99_latency_ms": 37.11,
 "stage_latency_breakdown_ms": {
   "preprocess_crop_resize": 0.42, "dinov2_vit_b14_frozen": 16.70,
   "spatial_projector_temporal_aux": 1.34,
   "laya_modernbert_e2e_traj": 14.70,
   "safety_shield_bicycle_mpc_pid": 0.45},
 "mean_cross_track_error_m": 0.085, "rms_steer_delta": 0.0292,
 "shield_interventions": {"rule1_emergency_brake": 51,
   "blindspot_left_occupied": 5}}
```


Tables 2–4 compare variants: without DINOv2+Laya (`rule_only`), the vehicle stalls behind lane blockages in `FSD-2, 5, 7, 8` (6/10 completed), whereas DINOv2+Laya + quintic S-curve + CTH-ACC achieves 10/10 completion and $\overline{|e_y|} = 0.085$ m (limitations: simulator state queries for baseline gating and fixed 50 ms synchronous step).

References

[1] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, Tianli Ding, Danny Driess, Avinava Dubey, Chelsea Finn, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv preprint arXiv:2307.15818*, 2023.

[2] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9650–9660, 2021.

[3] Dian Chen, Brady Zhou, Vladlen Koltun, and Philipp Krähenbühl. Learning by cheating. In *Conference on Robot Learning (CoRL)*, pages 66–75, 2020.

[4] Kashyap Chitta, Aditya Prakash, Bernhard Jaeger, Zehao Yu, Katrin Renz, and Andreas Geiger. TransFuser: Imitation with transformer-based sensor fusion for autonomous driving. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 45(11):12878–12895, 2022.

[5] Felipe Codevilla, Matthias Müller, Antonio López, Vladlen Koltun, and Alexey Dosovitskiy. End-to-end driving via conditional imitation learning. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 4693–4700, 2018.

[6] ConvAI Innovations. Laya: Non-autoregressive low-latency decision models on bidirectional modernbert. https://huggingface.co/convaiinnovations/laya, 2024.

[7] R. Craig Coulter. Implementation of the pure pursuit path tracking algorithm. Technical Report CMU-RI-

Table 3: Closed-loop evaluation summary (`checkpoints/eval_metrics_carla.json` and `eval_metrics.json`).

Simulator	Metric	Target KPI	Measured Result	Status
CARLA 0.9.16 (`Town10HD_Opt`)	Single-Frame Latency (Full FSD v2 / Base)	$\leq 40.0\,\text{ms}$ ($\geq 25\,\text{FPS}$)	33.40 ms (29.9 FPS, P95: 34.37 ms) / 31.67 ms	PASS
	Route Completion Rate (10 Scenarios, Seeds 5000–5009)	$\geq 85.0\%$	100.0% (10/10 episodes, 680.0 m, 1,986 frames)	PASS
	Crash-Free Rate (Vehicle / Pedestrian / Off-road)	$\geq 95.0\%$	100.0% (0 / 0 / 0)	PASS
	Validation ECE (Uncalibrated $\rightarrow$ Calibrated, $M = 15$)	≤ 0.0500	$0.0060 \rightarrow 0.0000$	PASS
	Scenario Composite Driving Score (DS_{scen})	$\geq 85.0\%$	100.0%	PASS
	Cross-Track Error $\overline{\lvert e_y \rvert}$ / Aux Depth MAE	$\leq 0.25\,\text{m}$ / $\leq 5.0\,\text{m}$	0.085 m / 3.36 m	PASS
MetaDrive 0.4.3 (50 Unseen Maps)	Single-Frame Latency (Seeds 5000–5049)	$\leq 40.0\,\text{ms}$ ($\geq 25\,\text{FPS}$)	30.20 ms (33.1 FPS, P95: 31.67 ms)	PASS
	Destination Arrival Rate (50 Episodes, 4,153 Frames)	$\geq 85.0\%$	100.0% (50/50 arrived; 98.7% arc progress)	PASS
	Crash-Free Rate	$\geq 95.0\%$	100.0%	PASS
	Calibrated ECE ($M = 15$ Bins)	≤ 0.0500	0.0000	PASS

Table 4: Model vs. controller ablation in CARLA 0.9.16 (10 scenarios).

System Variant	Completed	R_c	$\overline{\lvert e_y \rvert}$ (m)	$\lvert \Delta\delta \rvert_{\text{RMS}}$	AEB
Rule-Only + PID (No DINOv2/Laya)	6/10	76.4%	0.142	0.0195	58
Learned DINOv2+Laya (Step WPs)	10/10	100.0%	0.218	0.0642	138
+ Linear Steer Ramp $[\delta_{\text{from}} \rightarrow \delta_{\text{to}}]$	10/10	100.0%	0.135	0.0164	138
+ Quintic S-Curve & CTH-ACC	10/10	100.0%	0.092	0.0125	51
+ E2E 6-WP Traj, MPC & 360 BS (Full)	10/10	100.0%	0.085	0.0292	51

TR-92-01, Carnegie Mellon University, The Robotics Institute, 1992.

[8] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 4171–4186, 2019.

[9] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.

[10] Alexey Dosovitskiy, German Ros, Felipe Codevilla, Antonio Lopez, and Vladlen Koltun. CARLA: An open urban driving simulator. In *Proceedings of the 1st Annual Conference on Robot Learning (CoRL)*, pages 1–16, 2017.

[11] Marius Dupuis, Martin Strobl, and Hans Grezlikowski. OpenDRIVE format specification, rev. 1.4. *VIRES Simulationstechnologie GmbH*, 2010.

[12] Tamar Flash and Neville Hogan. The coordination of arm movements: An experimentally confirmed mathematical model. *Journal of Neuroscience*, 5(7):1688–1703, 1985.

[13] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning (ICML)*, pages 1321–1330, 2017.

[14] Peter E. Hart, Nils J. Nilsson, and Bertram Raphael. A formal basis for the heuristic determination of minimum cost paths. *IEEE Transactions on Systems Science and Cybernetics*, 4(2):100–107, 1968.

[15] Gabriel M. Hoffmann, Claire J. Tomlin, Michael Montemerlo, and Sebastian Thrun. Autonomous automobile trajectory tracking for off-road driving: Controller design, experimental validation and racing. *Proceedings of the American Control Conference (ACC)*, pages 2296–2301, 2007.

[16] Yihan Hu, Jiazhi Yang, Li Chen, Keyu Li, Chonghao Sima, Xizhou Zhu, Siqi Chai, Senyao Du, Tianwei Lin, Wenhai Wang, et al. Planning-oriented autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 17853–17862, 2023.

[17] Petros A. Ioannou and C. C. Chien. Autonomous intelligent cruise control. *IEEE Transactions on Vehicular Technology*, 42(4):657–672, 1993.

[18] Bernhard Jaeger, Kashyap Chitta, and Andreas Geiger. Hidden biases of end-to-end driving models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 8240–8249, 2023.

[19] Bo Jiang, Shaoyu Chen, Qing Xu, Bencheng Liao, Jiajie Chen, Helong Zhou, Qian Zhang, Wenyu Liu, Chang Huang, and Xinggang Wang. VAD: Vectorized scene representation for efficient autonomous driving. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 8340–8350, 2023.

[20] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, et al. OpenVLA: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024.

[21] Quanyi Li, Zhenghao Peng, Lan Feng, Qihang Zhang, Zhenghai Xue, and Bolei Zhou. MetaDrive: Composing diverse driving scenarios for generalizable reinforcement learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 45(3):3461–3475, 2022.

[22] Zhiqi Li, Wenhai Wang, Hongyang Li, Enze Xie, Chonghao Sima, Tong Lu, Yu Qiao, and Jifeng Dai. BEVFormer: Learning bird's-eye-view representation from multi-camera images via spatiotemporal transformers. In *European Conference on Computer Vision (ECCV)*, pages 1–18, 2022.

[23] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

[24] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

[25] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research (TMLR)*, 2024.

[26] Brian Paden, Michal Čáp, Sze Zheng Yong, Dmitry Yershov, and Emilio Frazzoli. A survey of motion planning and control techniques for self-driving urban vehicles. *IEEE Transactions on Intelligent Vehicles*, 1(1):33–55, 2016.

[27] Aditya Prakash, Kashyap Chitta, and Andreas Geiger. Multi-modal fusion transformer for end-to-end autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7077–7087, 2021.

[28] Rajesh Rajamani and Chuan Zhu. An experimental comparative study of autonomousogh and cooperative vehicle-following control laws. *IEEE Transactions on Control Systems Technology*, 10(3):371–383, 2002.

[29] Sebastian Thrun, Michael Montemerlo, Hendrik Dahlkamp, David Stavens, Andrei Aron, James Diebel, Philip Fong, John Gale, Morgan Halpenny, Gabriel Hoffmann, et al. Stanley: The robot that won the DARPA grand challenge. *Journal of Field Robotics*, 23(9):661–692, 2006.

[30] Martin Treiber, Ansgar Hennecke, and Dirk Helbing. Congested traffic states in empirical observations and microscopic simulations. *Physical Review E*, 62(2):1805–1824, 2000.

[31] Benjamin Warner, Antoine Chaffin, Benjamin Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher, Raja Biswas, Faisal Ladhak, Tom Aarsen, et al. Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. *arXiv preprint arXiv:2412.13663*, 2024.

[32] Moritz Werling, Julius Ziegler, Sören Kammel, and Sebastian Thrun. Optimal trajectory generation for dynamic street scenarios in a Frenet frame. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 987–993, 2010.